Day 29: PCA Python Library Implementation

PRODUCTION-READY SKLEARN PIPELINE & OUTPUT VALIDATION

1. Production Pipeline Optimization

While understanding manual matrix coordinates and eigenvector rotation forms the mathematical baseline, deploying production models requires optimized scikit-learn architectures. The `PCA` modules execute low-level singular value decomposition natively to scale down inputs with sub-second execution speeds.

> *Haath se matrix math seekhna dimaag ke liye zaroori tha, lekin real project mein hum scikit-learn library ka use karte hain taaki badi se badi calculations sirf ek line mein bina kisi jhanjhat ke turant execute ho jayein.*

2. Concise Execution Script

The structured script below executes the complete transformation sequence on our production dataset in its tightest algorithmic composition:

```
from sklearn.decomposition import PCA

# 1. Hamara asli data (Income aur Purchase)
data = [[4, 2], [8, 4]]

# 2. SEEDHE EK LINE ME SOLUTION (PCA lagakar data transform kiya)
pc1_values = PCA(n_components=1).fit_transform(data)

# Result print karna
print(f"Customer A ka PC1: {pc1_values[0][0]:.3f}")
print(f"Customer B ka PC1: {pc1_values[1][0]:.3f}")
```

3. Verification Log Analysis

Upon execution, the terminal pipes out the optimized component matrix weights:

```
Customer A ka PC1: 2.236
Customer B ka PC1: -2.236
```

4. Mathematical Connection (Why the values match)

The numeric magnitude 2.236 generated by Python matches our manual matrix logic. If we look closely at our mathematical derivation, the length of the vector was evaluated using square root calculations:

Manual Math Value:

$\sqrt{5} \approx 2.236$

The library code scales the coordinates exactly to this value. The scikit-learn engine simply inverts the signs depending on internal coordinate system alignment, while keeping the data alignment perfectly equal to our vertical matrix system:

$$\textbf{Manual Vector Form} = \begin{vmatrix} -2.235 \\ 2.235 \end{vmatrix} \quad \textbf{Python Library Form} = \begin{vmatrix} 2.236 \\ -2.236 \end{vmatrix}$$

> *Dhyan dein ki library code ka answer hamare hath wale math (√5) se ekdum exact match karta hai. Sklearn internally data coordinate axis ko flip kar sakta hai, isliye vertical matrix mein sign (+/-) alag ho sakte hain, lekin data ke phailao ka asar ekdum barabar rehta hai.*